

Phosphate plant -- Twin

Section report: Twin

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-12-31T23:00:00 (365 days, 8568 operating hours)
Product	Dap from 29.0 % P2O5 ore, 1721.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 8760 hourly intervals of 64 signals generated 2026-09-23T14:45:05.265 basis :screening_plant_model

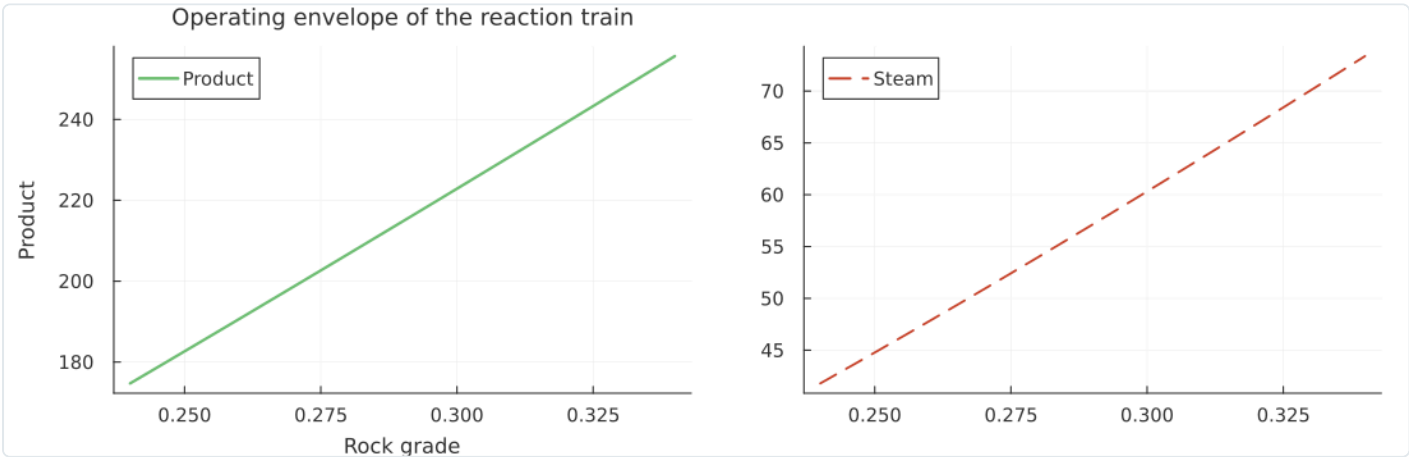
The digital twin: steady state, limits and alignment

The steady-state twin of the reaction train is a square system of 15 nonlinear balances solved with ModelingToolkit (ModelingToolkit nonlinear balance, mtkcompile + NonlinearProblem). It converged in 12.18 s with a largest residual of 0.0 -- the residual is reported because a solution nobody checked is a guess.

Unknown	Value	Unit	Description
:p2o5_fed	150	:t_ph	P2o5 fed
:excess	0.02562	:fraction	Excess
:free_so4	2.6606	:wt_pct	Free so4
:temperature	77.2418	:deg_c	Temperature
:efficiency	0.979187	:fraction	Efficiency
:p2o5_digested	146.878	:t_ph	P2o5 digested
:gypsum	737.2408	:t_ph	Gypsum
:p2o5_acid	142.4545	:t_ph	P2o5 acid
:water_liquor	366.0578	:t_ph	Water liquor
:acid_strength_weak	28.014	:wt_pct	Acid strength weak
:steam	65.1559	:t_ph	Steam
:water_evaporated	234.5613	:t_ph	Water evaporated
:ammonia	51.9605	:t_ph	Ammonia
:product	235.1041	:t_ph	Product
:merchant_p2o5	34.1891	:t_ph	Merchant p2o5

OPERATING WINDOW AT THE SOLUTION

Variable	At the solution	Low	High	Unit	Margin low	Margin high	Status
:acid_strength_weak	28.01	27	33	:wt_pct	1.01	4.99	Compliant
:ammonia	51.96	20	55	:t_ph	31.96	3.04	Compliant
:excess	0.0256	0.015	0.055	:fraction	0.0106	0.0294	Compliant
:free_so4	2.66	1.2	7	:wt_pct	1.46	4.34	Compliant
:steam	65.16	40	130	:t_ph	25.16	64.84	Compliant
:temperature	77.24	72	83	:deg_c	5.24	5.76	Compliant

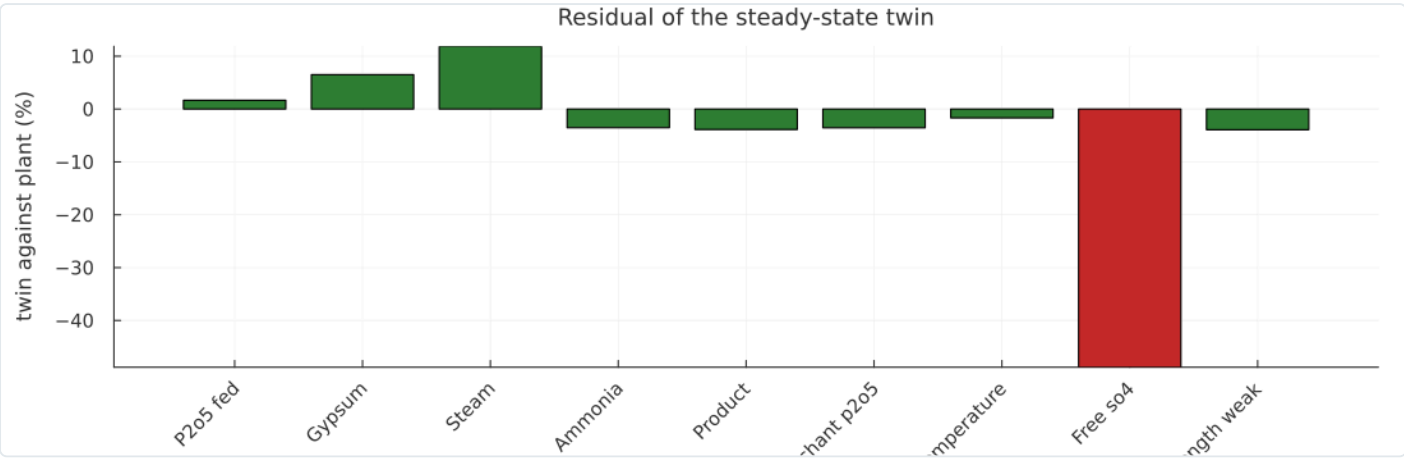


Operating envelope of the reaction train from the steady-state twin

Against the campaign it is supposed to describe, the twin scores 90.5 % of validity: the residuals below are what a commissioning would absorb as a bias, and the reason they are printed is that a twin is only as good as this table.

Variable	Twin	Plant	Unit	Residual %	Tolerance %	Status
:p2o5_fed	123.22	121.2	:t_ph	1.66	10	Compliant
:gypsum	603.24	566.39	:t_ph	6.5	12	Compliant
:steam	54.79	48.98	:t_ph	11.86	25	Compliant
:ammonia	42.67	44.23	:t_ph	-3.53	25	Compliant
:product	193.05	200.82	:t_ph	-3.87	25	Compliant
:merchant_p2o5	28.07	29.11	:t_ph	-3.55	20	Compliant

Variable	Twin	Plant	Unit	Residual %	Tolerance %	Status
temperature	76.43	77.73	:deg_c	-1.67	5	Compliant
free_so4	1.39	2.72	:wt_pct	-48.84	15	Noncompliant
acid_strength_weak	27.71	28.83	:wt_pct	-3.91	8	Compliant



Residual of the steady-state twin against the campaign it describes

The dynamic twin and the state estimation

The dynamic twin integrates 11 states with `OrdinaryDiffEqTsit5.Tsit5{typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!), FastBroadcast.Serial}(OrdinaryDiffEqCore.trivial_limiter!, OrdinaryDiffEqCore.trivial_limiter!, FastBroadcast.Serial())` on the hourly clock of the plant, from the same correlations the historian uses: the holdup of the attack tanks, the strength of the acid, the moisture of the granulator bed, the holdup of the mills and the fouling of the evaporator tubes.

DYNAMIC TRAJECTORY OF THE TWIN

Quantity	Value
Horizon	0.0 to 48.0 h
Status	compliant (Success)
P2O5 fed over the horizon	3,934.9 t
Liquor of the attack tanks	306 t to 306 t (-0.0009 %)
Highest temperature	78.02 C
Weakest acid	51.6 % P2O5
Fouling at the end	0.629
Integration time	4.81 s

The Kalman filter of the twin walked the historian for 336 intervals and estimated the states nobody measures. The innovation of every measurement is the table below: it is the residual the controller sees, and a sensor that drifts shows up here first.

Measurement	RMSE of the innovation	Bias	Intervals
:acid_strength	2.2	-2.12	336
:acid_strength_weak	40.48	40.1	336
:bed_moisture	1.41	1.4	336
:filter_rate	58.93	-58.79	336
:fouling_pct	0	0	336
:free_so4	27.28	-26.88	336
:granulator_temperature	1.06	-0.6231	336
:mill_inventory	142.25	-140.32	336
:mill_power	1,872.98	-1,846.22	336
:p80	51.53	-50.55	336
:product_rate	1,284.06	-1,281.39	336
:temperature	16.33	-0.9577	336

Generated by PHOSPHERE from the baseline and the seed recorded on the cover page.